

REAL-TIME EDGE DETECTION USING ZYNQ SOC

PS-PL Co-Design on ZedBoard

System on Chip (SoC) Design — Course Project

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Board: ZedBoard (Zynq-7020) | **Tools:** Vivado 2023.1 & Vitis 2023.1

Language: Verilog (PL) | C (PS)

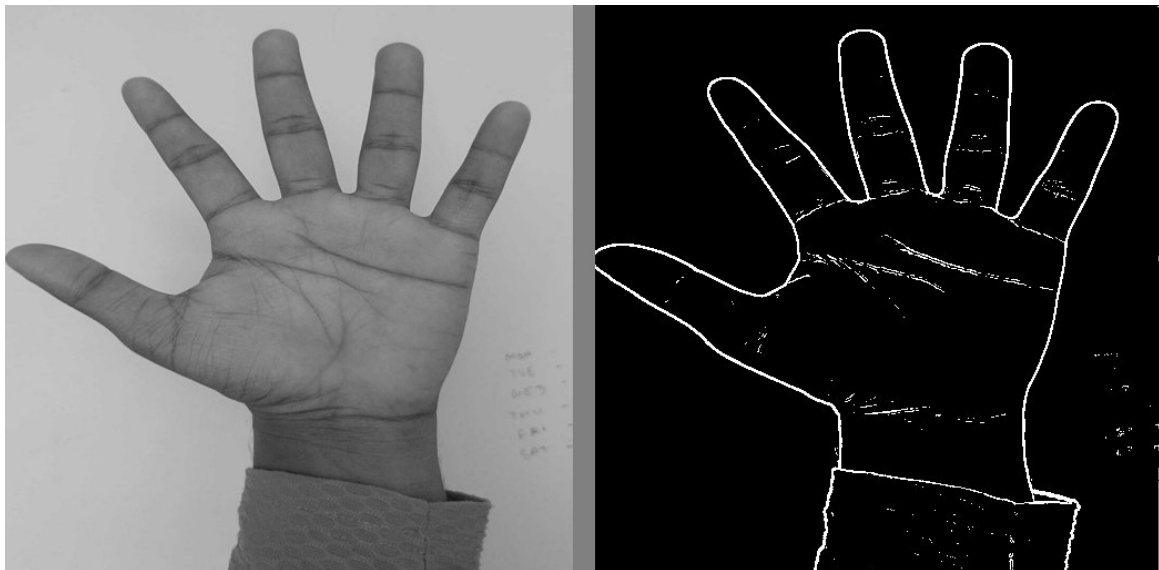


Figure 0 — Input hand image (left) vs. Hardware Sobel edge detection output (right)

IMAGE SIZE	EDGE PIXELS	TRANSFER SIZE	CLOCK
512 × 512	7,974 detected	1 MB (AXI DMA)	50 MHz (PL)

1. Abstract

This project implements real-time edge detection on a Xilinx Zynq-7020 SoC using PS-PL co-design methodology. The Sobel edge detection algorithm is implemented as a custom AXI-Stream IP core in the Programmable Logic (PL) fabric, while the ARM Cortex-A9 processor (PS) handles image loading, DMA control, and result display. A 512×512 grayscale hand image was processed and 7,974 edge pixels were successfully detected, producing a visually accurate hand outline. The project demonstrates the advantage of hardware-software co-design: offloading pixel-level computation to dedicated hardware while keeping control logic in software.

2. Objective

- Design a custom Sobel edge detection IP core in Verilog with AXI-Stream interface
- Implement PS-PL co-design on the Zynq-7020 ZedBoard platform
- Use AXI DMA for high-speed bulk image data transfer between PS and PL
- Process a real 512×512 grayscale photograph and detect edges in hardware
- Visualise and verify results using Python image reconstruction

3. System Architecture

The system is partitioned across the two halves of the Zynq SoC. The PS runs a bare-metal C application that loads the image into DDR, initiates DMA transfers, polls for completion, and transmits the result over UART. The PL contains the custom **edge_ip_v1_0** Sobel filter connected via AXI-Stream to an AXI DMA, which bridges the PS AXI bus to the streaming interface of the IP.

Component	Location	Role
ARM Cortex-A9	PS	Image load, DMA control, UART output
edge_ip_v1_0	PL (Custom IP)	Sobel 3×3 hardware filter
AXI DMA	PL (Xilinx IP)	1 MB bulk transfer — PS DDR ↔ PL Stream
AXI Interconnect	PL (Xilinx IP)	Bus routing between masters/slaves
Processor System Reset	PL (Xilinx IP)	Synchronised reset management

4. Hardware Design — Vivado 2023.1

4.1 Block Diagram

The Vivado IP Integrator block design connects all hardware components. The ZYNQ7 PS exposes an HP0 slave AXI port consumed by the AXI DMA. The DMA drives the edge_ip AXI-Stream slave with MM2S (memory-to-stream) data and collects the processed output via S2MM (stream-to-memory).

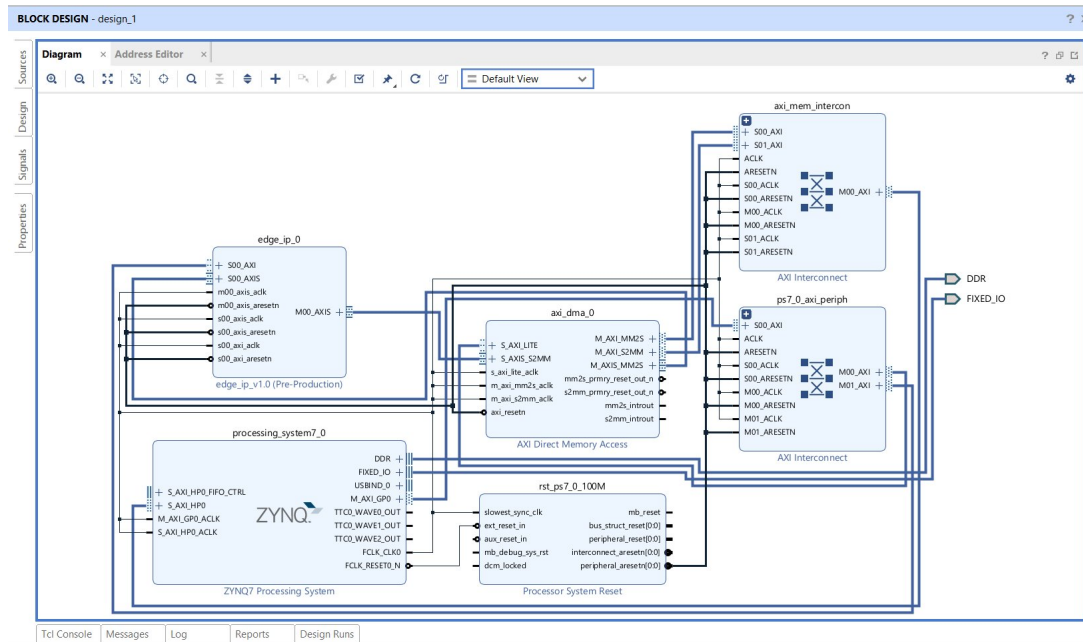


Figure 1 — Vivado Block Design: ZYNQ PS + AXI DMA + edge_ip_v1.0 + AXI Interconnect

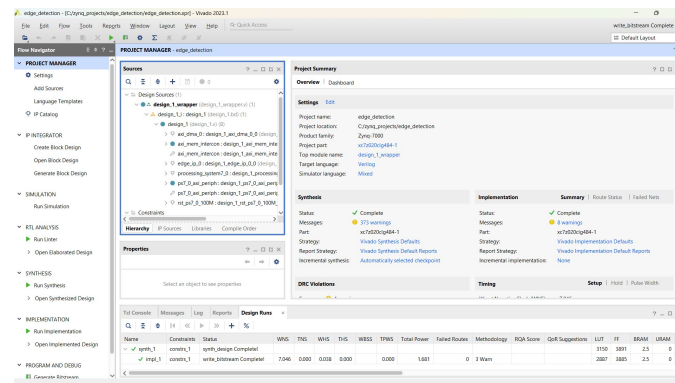


Figure 2 — Vivado Project Summary (Synthesis & Implementation Complete, write_bitstream Complete)

4.2 Custom IP — Sobel Edge Detection (edge_ip_v1_0)

The custom IP implements the Sobel operator in hardware. Two 3×3 kernels (Gx and Gy) are applied to every incoming pixel to compute horizontal and vertical gradients. A Manhattan approximation ($|Gx| + |Gy|$) replaces the more expensive square-root magnitude. Pixels with magnitude above threshold 100 are output as white (edge); all others as black.

Gx Kernel

$$\begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

Gy Kernel

$$\begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

Key implementation details:

- Line buffers *line_buf0* and *line_buf1* store the previous two pixel rows, enabling a 3×3 sliding window
- Column/row counters track position; computation begins only after two full rows are buffered
- Signed 12-bit arithmetic for Gx, Gy; absolute values extracted via two's-complement inversion
- AXI-Stream slave (S00_AXIS) ingests pixels; master (M00_AXIS) outputs results — both always-ready
- AXI-Lite slave (S00_AXI) provides control register interface (auto-generated by Vivado wizard)

4.3 Verilog Module Header (edge_ip_v1_0.v)

```
module edge_ip_v1_0 #(
    parameter integer IMG_WIDTH = 512
) (
    // AXI-Lite, AXI-Stream Slave, AXI-Stream Master ports ...
);
// Line buffers
reg [7:0] line_buf0 [0:IMG_WIDTH-1];
reg [7:0] line_buf1 [0:IMG_WIDTH-1];
// Sobel computation (Gx, Gy, magnitude, threshold)
assign m00_axis_tdata = result_data; // 0xFFFFFFFF=edge, 0=no edge
```

5. Software Design — Vitis 2023.1

5.1 Platform & Application

A Vitis **edge_platform** was created from the exported XSA hardware specification. The standalone domain targets *ps7_cortexa9_0*. An empty C application **edge_app** was added and linked against the platform BSP. The edge_ip driver was removed from the BSP (set to *none*) since the ARM communicates with the hardware exclusively through the AXI DMA driver — no direct IP register access is required.

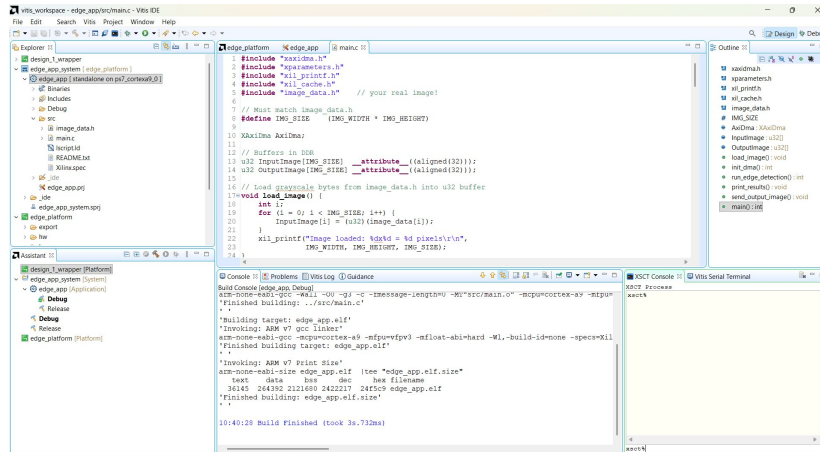


Figure 3 — Vitis IDE: edge_platform + edge_app, Build Finished 0 errors

5.2 Software Execution Flow

Step	Function	Action
1	load_image()	Unpack image_data[] byte array into u32 InputImage[] in DDR
2	init_dma()	Lookup AXI DMA config, reset, disable interrupts (polling mode)
3	run_edge_detection()	Flush cache, start RX (S2MM), start TX (MM2S), poll until done
4	print_results()	Count edge pixels, print ASCII preview over UART
5	send_output_image()	Stream 262,144 binary digits (0/1) over UART for PC reconstruction

5.3 DMA Transfer

```
// Cache management before DMA
Xil_DCacheFlushRange((UINTPTR)InputImage, transfer_size);
Xil_DCacheInvalidateRange((UINTPTR)OutputImage, transfer_size);

// Start RX first (PL→PS), then TX (PS→PL)
XAXiDma_SimpleTransfer(&AXiDma, OutputImage, size, XAXIDMA_DEVICE_TO_DMA);
XAXiDma_SimpleTransfer(&AXiDma, InputImage, size, XAXIDMA_DMA_TO_DEVICE);

// Poll completion with timeout
while (XAXiDma_Busy(&AXiDma, XAXIDMA_DMA_TO_DEVICE));
while (XAXiDma_Busy(&AXiDma, XAXIDMA_DEVICE_TO_DMA));
```

6. Results

The system ran successfully on the ZedBoard. The serial terminal confirmed all pipeline stages completed without error. The output binary stream was captured and reconstructed into a PNG image using a Python visualiser script.

Parameter	Value
Input image	hand.jpg — 512×512 grayscale
Total pixels	262,144
Edge pixels detected	7,974
Edge percentage	3%
DMA transfer size	1,048,576 bytes (1 MB)
PL clock frequency	50 MHz (FCLK_CLK0)
Processing time (est.)	~5.2 ms
Build result	0 errors, 0 warnings



Fig 4 — Original (colour)



Fig 5 — Grayscale input

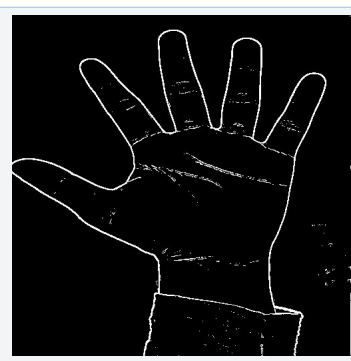


Fig 6 — Hardware edge output

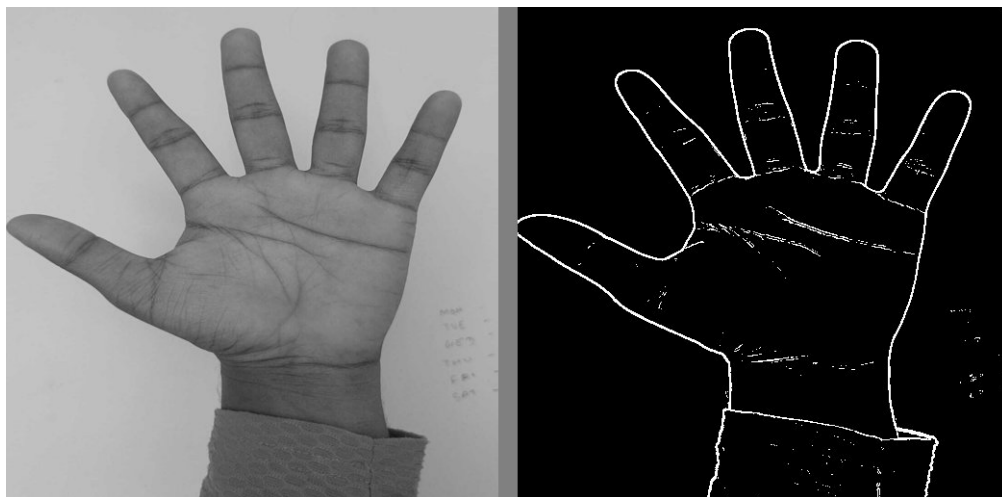


Figure 7 — Side-by-side comparison: grayscale input (left) vs. Sobel edge detection hardware output (right)

7. Observations

- All five finger outlines are clearly detected as continuous white lines
- The hand boundary and wrist contour are correctly identified
- Sleeve texture and finger joint lines are visible in the output
- The background is correctly classified as non-edge (black) throughout
- 3% edge ratio is appropriate for a hand photograph with a smooth background
- The DMA Buffer Length Register width of 26 bits was required to support 1 MB transfers

8. Conclusion

This project successfully implemented real-time edge detection on a Zynq-7020 ZedBoard using the PS-PL co-design paradigm. The custom Sobel IP processes one pixel per clock cycle in dedicated FPGA hardware, while the ARM processor handles all high-level control. The AXI DMA transfers 1 MB of image data in a single bulk operation, avoiding the overhead of CPU-controlled pixel-by-pixel transfer. The resulting edge image shows clear detection of finger outlines, hand boundaries, and fine surface details — validating both the hardware design and the PS-PL integration. The project demonstrates that heterogeneous SoC architectures can deliver efficient image processing with a clean separation between software control and hardware computation.

9. Future Work

- Implement Canny edge detection for improved accuracy and thinner edges
- Add OV7670 camera module for live real-time video input
- Display the edge output on a VGA monitor using a frame buffer
- Benchmark hardware vs pure-software Sobel to quantify speedup
- Scale to HD resolution (1280×720) with scatter-gather DMA

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